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Computer Vision
Experiment-2

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# Experiment No. 2
# Contrast Enhancement using
Python and OpenCV

import cv2
import numpy as np
import matplotlib.pyplot as plt

# Import libraries
print("Libraries imported
successfully.")

# Load image
image_path = "image.jpg"
img = cv2.imread(image_path)

if img is None:
    print("Error: Image not found.")
    exit()

# Convert to grayscale
gray = cv2.cvtColor(img,
cv2.COLOR_BGR2GRAY)

# Original image and histogram
plt.figure(figsize=(12, 5))
plt.subplot(1, 2, 1)
plt.imshow(gray, cmap="gray")
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.hist(gray.ravel(),
bins=256, range=[0, 256])
plt.title("Original
Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()

# Contrast stretching
min_pixel = np.min(gray)
max_pixel = np.max(gray)

if max_pixel != min_pixel:
    stretched = ((gray - min_pixel)
/
                (max_pixel - min_pixel) *
                255).astype(np.uint8)
else:
    stretched = gray.copy()

plt.figure(figsize=(10, 5))
plt.subplot(1, 2, 1)
plt.imshow(stretched, cmap="gray")
plt.title("Contrast Stretched")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.hist(stretched.ravel(),
bins=256, range=[0, 256])
plt.title("Stretched
Histogram")
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plt.xlabel("Intensity")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()

# Histogram equalization
equalized = cv2.equalizeHist(gray)
plt.figure(figsize=(10, 5))
plt.subplot(1, 2, 1)
plt.imshow(equalized, cmap="gray")
plt.title("Histogram Equalized")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.hist(equalized.ravel(),
bins=256, range=[0, 256])
plt.title("Equalized
Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()

# CLAHE
clahe = cv2.createCLAHE(
    clipLimit=2.0,
)
tileGr=(8,
8)
idSize

clahe_image = clahe.apply(gray)
plt.figure(figsize=(10, 5))
plt.subplot(1, 2, 1)
plt.imshow(clahe_image,
cmap="gray")
plt.title("CLAHE Enhanced")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.hist(clahe_image.ravel(),
bins=256, range=[0, 256])
plt.title("CLAHE Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()

# Image comparison
plt.figure(figsize=(15, 5))
plt.subplot(1, 3, 1)
plt.imshow(gray, cmap="gray")
plt.title("Original")
plt.axis("off")

plt.subplot(1, 3, 2)
plt.imshow(equalized, cmap="gray")
plt.title("Histogram
Equalization")
plt.axis("off")

plt.subplot(1, 3, 3)
plt.imshow(clahe_image,
cmap="gray")
plt.title("CLAHE")
plt.axis("off")

plt.tight_layout()

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plt.show()

# Histogram comparison
plt.figure(figsize=(12, 6))
plt.hist(gray.ravel(),
bins=256, range=[0, 256],
alpha=0.5,
label="Original")
plt.hist(equalized.ravel(),
bins=256, range=[0, 256],
alpha=0.5,
label="Equalized")
plt.hist(clahe_image.ravel(),
bins=256, range=[0, 256],
alpha=0.5, label="CLAHE")
plt.title("Histogram Comparison")
plt.xlabel("Intensity")
plt.ylabel("Frequency")
plt.legend()
plt.show()

# Save images
cv2.imwrite("contrast_stre
tched.jpg", stretched)
cv2.imwrite("histogram_equ
alized.jpg", equalized)
cv2.imwrite("clahe_enhance
d.jpg", clahe_image)
print("Enhanced images
saved successfully.") #
Analysis
print("\nImage Analysis:")
print("Original Minimum:",
np.min(gray))
print("Original Maximum:",
np.max(gray))
print("Stretched
Minimum:",
np.min(stretched))
print("Stretched
Maximum:",
np.max(stretched))
print("Equalized
Minimum:",
np.min(equalized))
print("Equalized
Maximum:",
np.max(equalized))
print("CLAHE Minimum:",
np.min(clahe_image))
print("CLAHE Maximum:",
np.max(clahe_image))
print("\nExperiment
completed
successfully.")

```

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Output :-

